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Lab 1: Requirements Engineering & UML Use-Case Modelling

PROBLEM STATEMENT #26 | Smart Cities, Transport & Logistics
Peer-to-Peer Carpooling & Expense Splitter

1. Problem Context & Overview

A commuter ridesharing portal that matches drivers and passengers with overlapping daily routes, enforces detour tolerance thresholds, and calculates automated fuel cost share splits.

Target Stakeholders / Actors: *Rider Commuter, Driver Host*

- **Functional Requirement (FR)**

Req id	type	Descrpition	priority	Acceptances Criteria	Rational (short)	Comments
FR-001	Functional	The system shall allow the **Rider Commuter to search or match rides** based on the required route	High	When the Rider searches for a route, the system displays available matching rides.	Helps the rider find a suitable carpool ride.	Related to **Search or Match Ride**.
FR-002	Functional	The system shall allow the **Driver Host to create a ride** by providing the required ride details.	High	When valid ride details are entered, the ride is created and made available for matching.	Allows drivers to offer rides to commuters.	Related to **Create Ride**.
FR-003	Functional	The system shall allow the **Rider Commuter to request a ride** from a matched ride.	High	When the rider selects a compatible ride and submits a request, the request is recorded.	Allows commuters to join a suitable ride.	Related to **Request Ride**.
FR-004	Functional	The system shall allow the **Driver Host to accept or reject a ride request**.	High	When the driver selects Accept or Reject, the request status is updated accordingly.	Allows the driver to control ride participation.	Related to **Accept or Reject Request**.

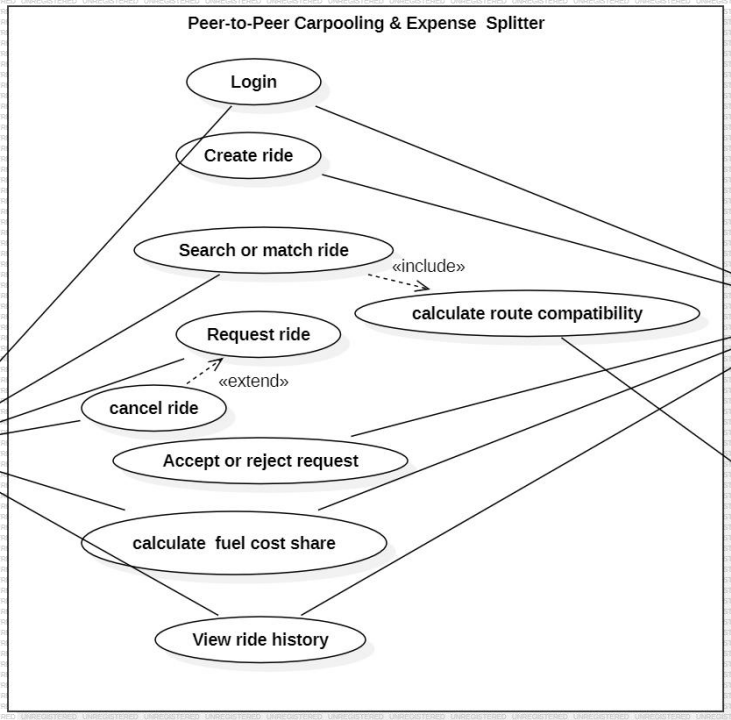
FR-005	Functional	The system shall **calculate the fuel cost share** for the participants of a carpool ride.	High	When the required ride and cost information is available, the system displays the fuel cost share.	Supports the expense-splitting purpose of the system.	Related to **Calculate Fuel Cost Share**.

- **NonFunctional Requirments (NFR)**

Req-no	Type	Description	Priority	Acceptances Criteria	Rational (short)	Comments
NFR-001	Non Functionla	The system shall **securely store user ratings, driving-license verification documents, and travel histories with encrypted backups**.	HIgh	A security check confirms that the specified data and backups are encrypted and protected from unauthorized access.	Protects sensitive user information.	Security requirement.
NFR-002	Non Functionl	The system shall provide **fast ride matching and route compatibility checking** under normal operating conditions.	HIgh	A normal ride search completes and displays the matching result within the specified system performance limit.	Provides a responsive experience for commuters.	Performance requirement.

- **UML Use-Case Diagram**





Use-Case Flow Document

- **Use Case: Request Ride**

Primary Actor: Rider Commuter
Supporting Actor: Driver Host, Route Service

- **Preconditions**
- The **Rider Commuter is logged in** to the system.
- The **Driver Host has created an available ride**.
- The Rider has entered the required route/search details.
- The Route Service is available to check route compatibility.

- **Main Success Scenario**
- The Rider searches for a ride by entering the required route.
- The system displays available rides.
- The system checks the route compatibility between the Rider and Driver.
- The **Route Service** calculates the route and detour distance.
- The system verifies that the detour is within the allowed **1.5 km** limit.
- The system displays the compatible ride to the Rider.

- The Rider selects the required ride.
- The Rider submits a **ride request**.
- The system records the request and sends it to the **Driver Host**.
- The Driver Host reviews the request.
- The Driver Host accepts the request.
- The system updates the request status to **Accepted**.
- The Rider can view the confirmed ride.

- **Alternate Flow 1 — Detour Exceeds 1.5 km**
- During route compatibility checking, the Route Service calculates a detour greater than **1.5 km**.
- The system identifies the ride as incompatible.
- The system does not allow the Rider to request that ride.
- The system displays a message that the ride exceeds the allowed detour limit.
- The Rider can search for another ride.

Alternate Flow 2 — Driver Rejects Request

- The Driver Host receives the Rider's request.
- The Driver reviews the request and selects **Reject**.
- The system changes the request status to **Rejected**.
- The Rider is informed that the request was rejected.
- The Rider can search for another available ride.

- **Postconditions**
- **If successful:**

  - The ride request is stored with **Accepted** status.
  - The Rider is associated with the selected ride.
  - The confirmed ride can be viewed in ride history.

- **If unsuccessful:**

  - No request is created for an incompatible ride.
  - The Rider can search for another ride.